Absolute sky radiance through deep twilight: a backward Monte Carlo model with three-dimensional cloud transport, externally validated to solar zenith angle 103 degrees

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Abstract

Deep twilight, solar zenith angles (SZA) between roughly 96° and 108°, is a difficult regime for radiative transfer: the line of sight lies entirely inside the Earth's shadow, single scattering vanishes, and the observable radiance is carried by multiply scattered light that has crossed the terminator at high altitude. We present twilight, an open-source backward Monte Carlo model that computes absolute spectral sky radiance through this regime in a spherically stratified, refracting atmosphere extending to 150 km, at 41 wavelengths with full Stokes polarization, and with measured three-dimensional cloud fields traversed voxel by voxel. Variance reduction combines forced collisions implemented as truncated null-collision tracking against per-shell majorants, collision-location importance sampling, directionally biased escape sampling of the Dwivedi form, weight-window population control driven by a deterministic solar-source importance map, hero-wavelength spectral multiple importance sampling, and a bidirectional connection estimator that couples every backward-chain collision to light subpaths traced from the sunlit terminator, covering each scattering order beyond the second from the sunlit side; each component is provably unbiased and the implementation is held to that standard by analytic unit gates, bit-identity harnesses, and statistical parity tests between four independent implementations (scalar and polarized CPU chains, and Metal and WGSL GPU kernels). Validation proceeds in three rings. Against reference codes: absolute clear-sky radiance agrees with pseu-

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dospherical DISORT to ratios of 0.975 to 0.993 at SZA 60° to 85°; 144 of 144 cloud-slab comparison points agree with DISORT and MYSTIC within 3% plus two standard errors; true-3D cloud cubes agree with SHDOM in 112 of 112 points; and cloud decks at twilight geometry are refereed against MYSTIC runs of up to 10^9 photons, to SZA 103°, with every cell of the tier, one-dimensional and three-dimensional cloud paths alike, statistically consistent with its reference (ratios 0.87 to 1.19). Against measurements: the simulated zenith twilight decay reproduces published sky-brightness data to 1.2 to 5.5% per magnitude of decay. Against human observation: a psychophysical detection layer, calibrated once against desert campaigns reporting a documented central-mean dawn, reproduces the dawn detection times of sixteen published visual and instrumental campaigns with a median absolute residual of 0.26° of solar depression over the fourteen scored rows, including an out-of-sample seasonal sweep at 52°N. The model, its validation caches, and every table in this paper are reproducible from the public repository.

Keywords: radiative transfer, Monte Carlo, twilight, sky brightness, polarization, three-dimensional clouds, visual detection

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1. Introduction

The brightness of the twilight sky is one of the oldest quantitative problems in atmospheric optics [29], and it remains one of the least served by modern radiative transfer tools. The reason is geometric. Once the Sun is more than about 6° below the horizon, the observer's entire line of sight lies inside the Earth's shadow; by 12° the shadow height above the observer exceeds the depth of the dense atmosphere, and essentially no singly scattered photon can reach the eye. What remains is light that was scattered at high altitude on the sunlit side of the terminator, transported laterally over hundreds of kilometers, and scattered again, possibly several times, into the line of sight. Plane-parallel methods are structurally unable to represent this transport; pseudospherical corrections extend them usefully to solar zenith angles (SZA) near 90° to 95° but not beyond. Fully spherical Monte Carlo codes such as MYSTIC [23, 9] can in principle go deeper, but published validation and practical convergence reach SZA of roughly 100°; past that point the reference codes themselves become the bottleneck.

Yet the deep-twilight regime is where several physically interesting and socially important observables reside: the visual onset of dawn and the disappearance of dusk colors (the basis of prayer timekeeping for roughly two billion people), the visibility of the zodiacal light and its false dawn, astronomical twilight for observatory scheduling, and the photobiology of natural light exposure at high latitudes. All of these are questions about absolute radiance in a regime where absolute radiance is hard to compute and hard to validate.

This paper describes twilight, an open-source simulator built for this regime, and the validation program that establishes what it can and cannot currently claim. Three design decisions distinguish it from general-purpose radiative transfer packages. First, the transport is backward (adjoint) Monte Carlo from the observer, in a spherically stratified, refracting atmosphere, with the entire sampling budget concentrated on the multiply scattered field: the first scattering order is evaluated as an exact deterministic integral, and the Monte Carlo estimator begins at order two. Second, clouds are treated as measured three-dimensional fields, not homogeneous layers: a georeferenced voxel volume, reconstructed from geostationary satellite imagery by a neural network trained on spaceborne cloud radar [3], is traversed exactly by the photon chains. Third, the estimator stack was engineered specifically against the heavy-tailed variance that dominates deep twilight, where the observable mean is carried by rare chains that climb to the sunlit terminator; we document both the machinery and its measured effect, including the explicitly reported cases where affordable budgets remain statistically weak.

A further methodological point deserves emphasis. Monte Carlo transport codes fail most dangerously not by crashing but by converging confidently to a biased answer; the literature of radiative transfer intercomparisons is in large part a record of such discoveries. We therefore treat the verification process itself as part of the model. Every estimator component carries an analytic unit gate; every refactoring is pinned by bit-identity harnesses over the unchanged surfaces; four independent implementations (a scalar CPU chain, a polarized CPU chain, a Metal GPU kernel, and a portable WGSL GPU kernel) are held in statistical parity with pre-registered tolerance bands derived from measured seed variance; and the code has twice been subjected to an adversarial review in which independent re-derivations of the estimator mathematics hunted specifically for compensating-error pairs. Findings from these reviews, sixteen in the most recent round, all closed, are recorded in the repository together with their fixes.

To state the contributions plainly: to our knowledge this work provides (i) a bidirectional connection estimator for the deep-twilight regime, coupling every backward-chain collision beyond the terminator to light subpaths traced from the sunlit top of atmosphere through an exact per-path balance-heuristic MIS, which is what makes the regime tractable at affordable budgets; (ii) exact voxel-by-voxel transport of measured three-dimensional cloud fields at twilight geometry, where the light path crosses the cloud field hundreds of kilometres from the observer and a horizontally uniform deck is not a usable approximation; (iii) the deepest external radiative transfer comparison attempted for cloud decks at twilight geometry (solar zenith angle 103° against 10^9-photon MYSTIC references, with every cell of the deep tier, one-dimensional and three-dimensional field paths alike, gated and statistically consistent; published comparisons stop near 100° under clear skies); and (iv) the first twilight model validated simultaneously against absolute pseudo-spherical reference codes, measured sky brightness, and a century of human dawn observation.

We are deliberate about what the last of these can and cannot claim. The deep comparison establishes agreement at the tens-of-percent level, because the acceptance bands there are 25 to 46 percent wide and are dominated by the *referee's* own Monte Carlo error rather than the model's (measured across the tier: referee relative standard error 5.9 to 10.7 percent against the model's 2.6 to 5.9, one field cell excepted where the two are level at 9.9, so the referee is the larger term in fifteen of sixteen cells). This is a weaker statement than the percent-level DISORT anchor at moderate solar zenith angle, and it is weaker because past 100° the reference codes, not the model, are the limiting instrument. Section 5 reports the decomposition rather than the pass count alone.

The engineering practice that supports these results – four parity-locked implementations, pre-registered acceptance bands, bit-identity harnesses across refactors – is described in Section 4 and Section 5. We report it as methodology rather than as a result: agreement among four implementations of the same physics constrains implementation error, not physical correctness, and only the external referees and the measured skies can speak to the latter.

Section 2 describes the forward model: atmosphere, scattering physics, background light sources, and the transport estimator. Section 3 describes the variance-reduction stack and its measured effect. Section 4 describes the GPU implementations and the cross-implementation parity methodology. Section 5 presents the three-ring validation program: reference codes, measured skies, and human observation campaigns. Section 6 describes the propagation of input uncertainties to the output times. Section 7 states the current limitations plainly, and Section 8 concludes.

2. The forward model

2.1. Atmosphere and optical properties

The atmosphere is represented by 56 concentric spherical shells extending to 150 km, built on the US Standard Atmosphere 1976 with an extended thermosphere. Shell boundaries are concentrated where the twilight transport needs them: 1 km spacing in the troposphere and lower stratosphere, coarsening upward. Molecular (Rayleigh) scattering follows Bodhaine et al. [5]. Absorption includes five species: ozone (Serdyuchenko cross sections, 30), NO_2, the O_2 bands, water vapor, and the O_4 collision-induced continuum, with line data drawn from HITRAN [14] and pre-integrated onto the model's spectral grid. Aerosols follow the OPAC climatological types [17], scaled by a column optical depth that can be supplied from measurement (Section 6). The spectral grid is 41 wavelengths from 380 to 780 nm, sufficient to resolve the Chappuis band of ozone, whose absorption shapes twilight color.

Refraction is applied at every shell boundary by Snell's law on the local radial gradient of refractive index; rays bend continuously in the piecewise-constant-index limit, and total internal reflection (ducting) at grazing incidence is handled explicitly. Refraction matters twice in this regime: it lifts the apparent terminator, and it bends the long shadow-ray paths that decide whether a scattering event is illuminated.

Clouds enter in two forms. A one-dimensional form treats a deck as a gray extinction profile assigned to shells, with delta-Eddington scaling of the optical depth and asymmetry parameter. The three-dimensional form is a georeferenced voxel field of extinction and asymmetry (built from the satellite reconstruction of 3); chains traverse it with an exact digital-differential-analyzer walk in the style of Amanatides and Woo [2] on the latitude, longitude, altitude grid, and a two-level macrocell acceleration structure provides tile maxima for majorant construction (Section 3.1).

2.2. Background light sources

At the depressions where dawn detection occurs, the twilight excess being detected is comparable to, or below, the ambient night-sky radiance, so the background must be modeled absolutely and per direction. The model sums, for every simulated sky patch: zodiacal light from the measured Leinert tables [21] with the observer's heliocentric geometry; integrated starlight from the

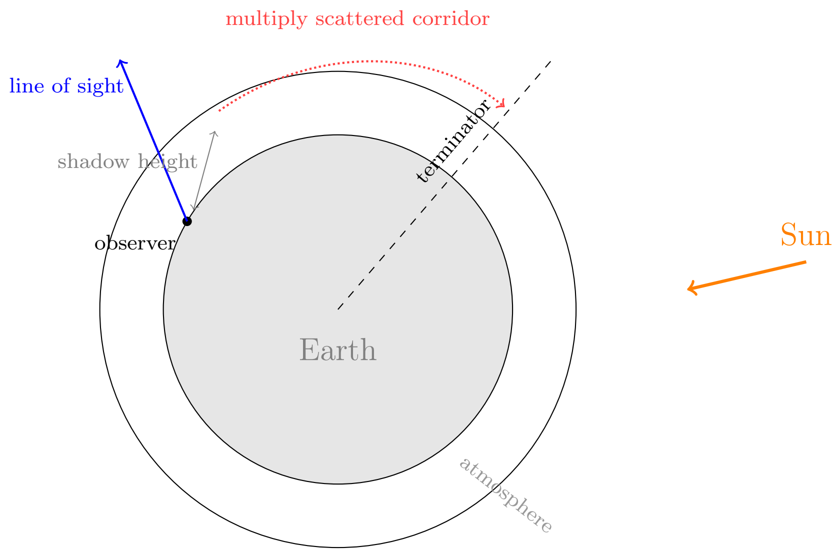

Figure 1: The deep-twilight geometry: beyond SZA $\sim 96°$ the observer's entire line of sight lies inside the Earth's shadow; the observable radiance is carried by photons scattered at high altitude on the sunlit side of the terminator, transported laterally, and scattered again into the line of sight (backward chains must discover this corridor in reverse). Every variance-reduction component of Section 3 exists to sample this corridor efficiently without bias.

Pioneer 10/11 photometry [32]; airglow driven by the measured F10.7 solar radio flux; moonlight following Krisciunas and Schaefer [20], supplied with the true lunar state (parallax, phase angle, distance) from the JPL DE440 ephemeris [26]; and artificial skyglow from the propagated world atlas [11], in its 2024 update, cross-checked against Visible Infrared Imaging Radiometer Suite (VIIRS) nighttime lights, propagated to the viewing geometry with a Garstang-type model [13]. The artificial veil is applied in mesopic photometry computed from the source spectrum (high-pressure sodium versus LED mixtures), because a fixed photopic conversion over-weights sodium-dominated cities by more than a factor of two at threshold adaptation levels [6].

2.3. Solar geometry

Solar position uses the NREL solar position algorithm (SPA) [28]; when the DE440 kernel is available the Sun and Moon states are taken from the ephemeris directly. The pipeline scans SZA rather than clock time and converts the detected crossings back to time through the local solar motion, which decouples the transport cache from date and longitude.

2.4. Transport: exact first order plus backward Monte Carlo

Radiance along the line of sight is split as $L = L_1 + L_{2+}$. The single-scattering term L_1 is computed as a deterministic integral along the refracted line of sight: at each quadrature point the direct solar transmission is evaluated by a refracted shadow ray through the shell stack (with the cloud field, when present, attenuating by Beer-Lambert extinction along the exact voxel path), and the local phase matrix maps the solar beam into the view direction. This term is exact up to quadrature and dominates civil twilight.

The multiply scattered term L_{2+} is estimated by backward Monte Carlo. The line of sight is discretized into steps; from each step, secondary chains are seeded by a three-component directional mixture (phase-function proposal toward the Sun, a zenith-biased power-cosine lobe, and a terminator-oriented lobe), extended by a fourth lateral-escape lobe that activates only for cloud-coupled seeds under broken fields in the deep regime (Section 7), combined with a one-sample multiple importance sampling (MIS) weight in the balance heuristic [33], so the seed estimator is unbiased for any mixture. Chains then propagate by free-path sampling through the shell stack, scattering off the local Rayleigh-plus-Henyey-Greenstein mixture, with polarization tracked by Stokes vectors and the scattering-plane rotations applied in a fused form that avoids explicit Mueller matrix products. At every collision a next-event estimation shadow ray connects the chain to the Sun through the refracting shells and the cloud field. Ground interaction is Lambertian with configurable albedo, and ground-bounce next-event estimation is included.

Under a three-dimensional cloud field the chain's free path is sampled against a combined gas-plus-cloud channel. Because the voxel extinction is not constant within a shell, the inversion uses null-collision (Woodcock) tracking [12] against per-shell majorants: tile maxima from the macrocell structure, joined with the background column and rounded upward so that single-precision domination is guaranteed. Real collisions are classified gas versus cloud by the local extinction ratio; cloud vertices scatter from the local delta-scaled asymmetry as a depolarizing Henyey-Greenstein event. An equivalence gate (Section 5.1) verifies that a uniform field reproduces the one-dimensional deck chains exactly.

Figure 2 shows the polarization output directly: the degree of polarization of the multiply scattered zenith radiance across the visible grid, carrying the

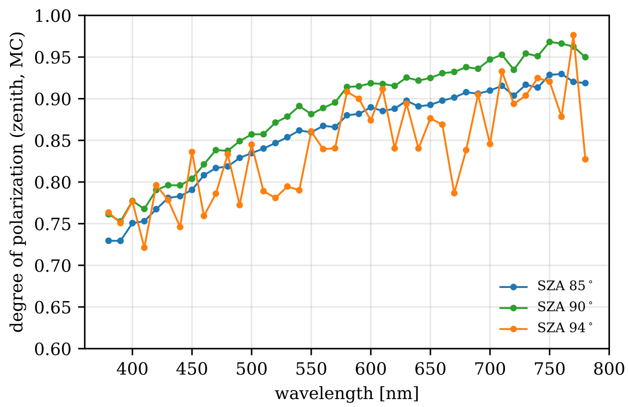

Figure 2: The degree of polarization of the backward Monte Carlo zenith radiance is shown across the spectral grid at three SZA values (clear sky, 40 000 chains per wavelength). Deeper twilight requires substantially larger budgets for per-wavelength polarization convergence and is omitted.

expected rise with wavelength as the Rayleigh single-scattering character strengthens.

3. Variance reduction for the deep-twilight tail

At SZA beyond about 100° the estimator confronts a specific pathology: almost all chains die in the dark, and the mean is carried by rare chains that random-walk upward and sunward until they reach illuminated altitudes. Left untreated, this produces heavy-tailed seed distributions whose sample variance understates the true error, the most dangerous failure mode a Monte Carlo code has, because it looks like convergence. The stack described below was built against this failure mode. Each component is unbiased by construction; the telescoping identities are stated in the code alongside the implementation and exercised by analytic unit gates.

3.1. Forced collisions as truncated null-collision tracking

Beyond SZA 96° the free-path sampling is replaced, whenever the path to the atmosphere boundary is optically thin, by a forced collision: the chain weight is multiplied by $1 - e^{-\tau_{\max}}$ and the collision optical depth is drawn from the truncated exponential on $[0, \tau_{\max}]$. Under a cloud field, $\tau_{\max}$ is accumulated against the majorant-combined channel, and the forced stage

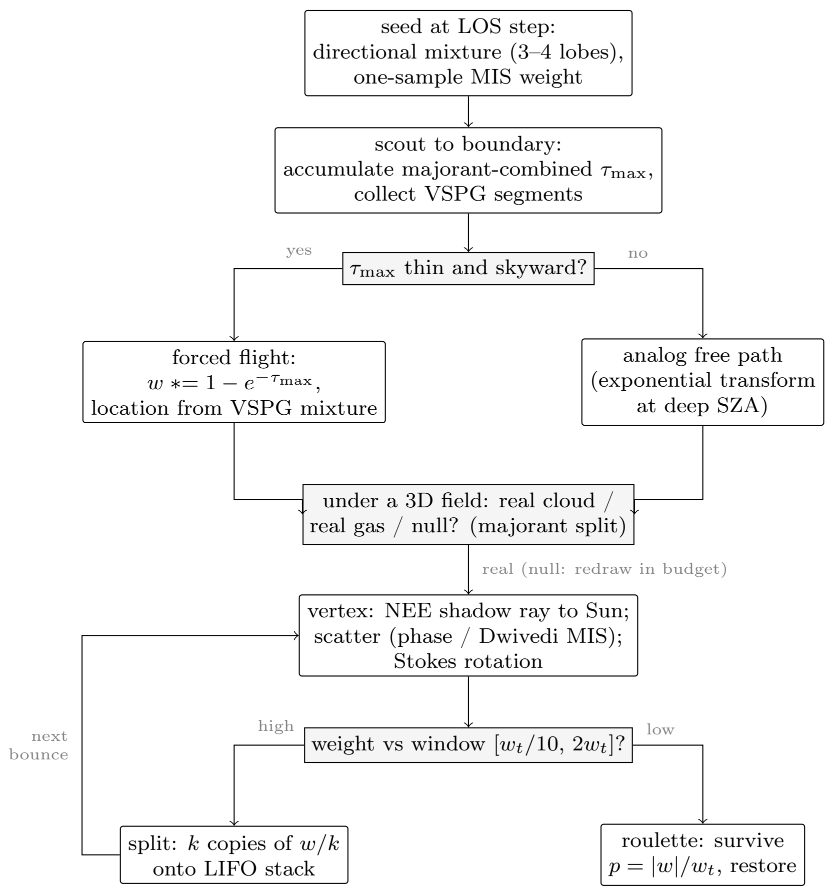

Figure 3: The lifecycle of one secondary chain, showing where each variance-reduction component of this section acts. Every branch is an unbiased identity: forced flights and VSPG reweight the free-path density, the null-collision loop telescopes to the analog first-arrival law, the directional MIS uses the balance heuristic, and splitting and roulette preserve expected weight.

composes with the null-collision classification: a drawn collision is real-cloud, real-gas, or null in proportion to the local split $\sigma_c(\mathbf{x}) : \sigma_g : (\bar{\sigma} - \sigma_c(\mathbf{x}))$, and null events redraw within the remaining truncated budget. The product of stage normalizers telescopes so that the sampled first-arrival law is exactly the analog one; the identity is verified by a 400 000-flight gate against the analytic first-collision density on a checkerboard field, under both tight and deliberately inflated ($\times 3$) majorants.

3.2. Collision-location importance (VSPG)

VSPG (volume scattering probability guiding) draws the collision location of a forced flight from an importance mixture rather than the bare truncated exponential.

Within a forced flight, the collision location is drawn not from the bare truncated exponential but from a piecewise-exponential importance mixture over the traversed shell segments, weighted by a calibrated function of segment altitude and SZA that concentrates collisions where the Sun still illuminates. The per-segment weights are collected during the same shell walk that accumulates $\tau_{\max}$, at no additional traversal cost. Buffer overflow on pathological multi-reflection paths extends the final segment so the segment set always tiles $[0, \tau_{\max}]$, preserving the exact normalization (an earlier silent truncation of this buffer, which redistributed collision probability toward the path head, is one of the compensating-error findings of the adversarial review).

3.3. Directional biasing (Dwivedi) with balance-heuristic MIS

Deep chains must travel laterally; an isotropizing phase function makes lateral progress diffusive and slow. Following the deep-penetration tradition of Dwivedi [8], a fraction of direction draws (ramped in with SZA) samples a horizontally biased distribution in the local frame, combined with the physical phase function through one-sample multiple importance sampling so the estimator remains exactly unbiased. In the polarized chain the balance-heuristic weight corrects the scalar intensity; the Stokes rotation uses the actually sampled angle, so polarization is transported without approximation.

3.4. Weight windows, splitting, and a solar-source importance map

Population control follows the weight-window formulation of the deep-penetration literature [34]: at each bounce the chain weight is compared with a target derived from an importance function of position; overweight chains split (each carrying weight w/k), underweight chains face Russian roulette that restores survivors to the target. Both operations are unbiased identities. The importance function is supplied either by a calibrated altitude-plus-lateral heuristic or by a deterministic solar-source map: in a spherically stratified atmosphere the solar in-scatter density depends only on altitude and the cosine of the position angle to the Sun, so a single 40×64 map, built per radiance call from the local scattering coefficient times the ground-shadow-aware

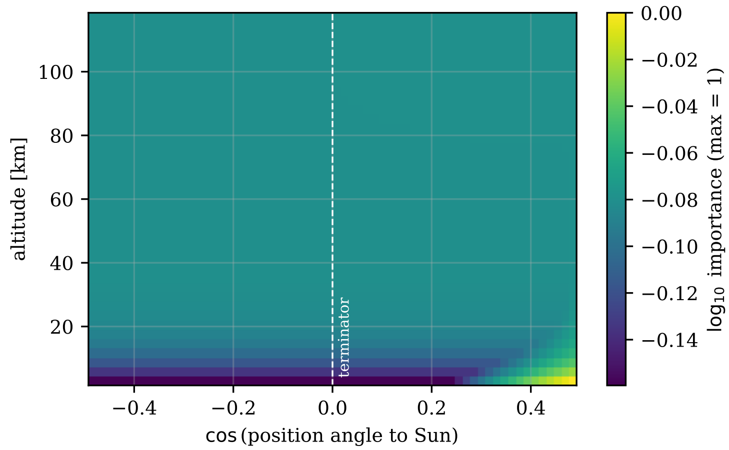

Figure 4: The deterministic solar-source importance map over altitude and the cosine of the position angle to the Sun, on the production clear-sky atmosphere at 450 nm (log scale, maximum normalized to one; the dashed line marks the terminator). The map saturates near unity through most of the domain because the upper atmosphere is nearly transparent; the residual structure (about 30%) sits in the low-altitude deep shadow. This saturation explains the measured result that weight windows alone do not reduce deep-twilight variance: survival weighting has nothing to act on when almost every position retains access to the sunlit source.

solar transmittance and relaxed by a Bellman-Ford transmittance recursion, serves every observer geometry. In measured ledgers the map matches the tuned heuristic (evidence that the heuristic's constants were near-optimal); it replaces hand tuning with physics and is now the default. Figure 4 shows the map, and its near-uniformity is itself a finding: above the boundary layer the atmosphere is sufficiently transparent that nearly every cell in the domain can reach sunlight at low optical cost, so the maximum-transmittance importance saturates within roughly 30% of unity everywhere except the deepest low-altitude shadow. Population control keyed to any such importance is therefore nearly inert in the deep regime, which is precisely what the variance ledger measured (weight windows alone left the deep-cell coefficient of variation unchanged); the binding constraints are the collision-location and directional distributions, not survival weighting.

3.5. Additional components

An exponential-transform path stretch biases free paths upward at deep SZA with the standard weight correction. Spectrally, chains are traced at a hero wavelength with adjoint-line-integration across the 41-wavelength grid: one geometric path is sampled using the hero wavelength's optical properties, and its contribution is carried to every other grid wavelength simultaneously by the ratio of per-wavelength path transmittances to the hero's (exact, because the path geometry is shared and only the extinction along it differs), so a single walk services the full spectrum rather than 41 monochromatic solves; the hero assignment rotates across chains and the estimates combine by spectral multiple importance sampling in the style of Wilkie et al. [35]; under three-dimensional fields the per-wavelength chains run individually (the per-wavelength null ratios under a shared majorant would otherwise be heavy-tailed, a documented design deferral). A bidirectional subpath reuse scheme contributes additional order-two connections on the main chain. In the deep regime this generalizes to every higher order: light subpaths traced from the sunlit top of the atmosphere maintain a registry of 512 importance-weighted scatter vertices, and beyond SZA 99° every collision vertex of the backward chain additionally makes a bidirectional connection to one registry vertex, so that each scattering order of three and above receives a contribution constructed from the sunlit side. At each order the connection is combined with its next-event counterpart through a per-path balance-heuristic multiple importance sampling weight; the two weights are evaluated by one shared function and sum to unity, so the combined estimator is unbiased by construction, and the connection scores are normalized by the registry pass attempt count, not by the count of successful connections. The scheme runs identically on the scalar hero-wavelength chain and the polarized Stokes chain, and a partition cross-check against the pre-connection estimator agrees to 2.3% at SZA 99.5°. Its effect is to remove the dependence of the deep signal on rare long lateral random walks; this is what resolves the one-dimensional SZA 103° referee cells of Section 3.6.

3.6. Measured effect

Table 1 reports the production estimator against the MYSTIC references (standard errors 5.9 to 10.7%) for the hardest configuration we can referee externally: a uniform deck of delta-scaled optical depth $\tau^* = 3$ under twilight geometry, at 1024 seeds of 16 000 photons per wavelength (an eight-fold increase over an earlier 128-seed pass). The regime is heavy-tailed. At this budget the referee gates all six cells, and the estimator is consistent with the reference in every one (0.87 to 1.13×, bands 0.28 to 0.40); an apparent forty-percent-low reading of the 550 nm SZA 101° cell at 128 seeds proved to be seed undersampling of the heavy tail, not a bias, and resolved on convergence. The three SZA 103° cells, which under the purely unidirectional estimator stayed heavy-tailed enough that the confidence band remained above half the reference, gate once the connection estimator of Section 3.5 supplies each scattering order of three and above from the sunlit side: across both deck optical depths all six one-dimensional SZA 103° cells pass, with bands 0.31 to 0.40 of the reference and ratios 0.93 to 1.13. The two three-dimensional-field cells at SZA 103°, which at their earlier 128-seed budget were reported low-power (consistent but not constraining), close the same way the 128-seed deck cells did, by seed statistics alone: at 1024 seeds ($\tau^* = 1$) and 512 seeds ($\tau^* = 3$) of 16 000 photons the referee gates both, with ratios 1.19 and 0.99 and bands 0.36 and 0.46 of the reference. The connection estimator requires the one-dimensional combined channel and deliberately leaves the field path untouched; the field cells were closed with the field estimator unchanged, so the earlier low-power verdicts were a statement about seed budget, not about the estimator. The reference side was itself replicated: eight fresh-seed MYSTIC replicas of the four $\tau^* = 3$ references (two per reference, at each reference's own budget of 3×10^8 or 10^9 photons) reproduce the cached values within their reported errors, except the 650 nm SZA 103° case, whose cached value sits 2.8 combined standard deviations below the pooled value of its replicas (a low draw); the model passes against both the cached and the replica-pooled references. On a synthetic broken-cloud field the same machinery drives a large recovery of mean radiance relative to the analog estimator in the deep cells where zenith starvation otherwise biases it low, with the seed coefficient of variation reduced correspondingly; the per-configuration before-and-after values are recorded in the archived field ledgers. On clear sky at the dawn-detection depressions the seed coefficient of variation improves by factors of up to five (median near two; the shallowest red cell is essentially unchanged at 1.05).

Figure 5 shows the ledger cell by cell, and Figure 6 places every referee cell with its error bar. Figure 7 shows the estimator's convergence across budgets: at moderate depression the relative standard error follows the $N^{-1/2}$ law from small budgets, while the deep cells show the heavy-tail regime as departures from the ideal slope.

Two observations from this campaign generalize beyond our code. First, the analog estimator (the before bars of Figure 5) illustrates the false-precision failure: it returned 5 to 26% of the true radiance with deceptively tight seed spread, because no seed contained the rare compensating chains. Sample

Table 1: Production estimator versus dedicated MYSTIC referees of 3×10^8 (SZA 101°) and 10^9 (SZA 103°) photons. Uniform cloud deck, delta-scaled optical depth $\tau^* = 3$, zenith view, 1024 seeds of 16 000 photons per wavelength. Ratios are model/referee; uncertainties are seed standard errors. The referee gates a cell only when the band $3\sqrt{\mathrm{se}^2 + \mathrm{se}_{\mathrm{ref}}^2} + 0.05\,r$ falls below half the reference r; cells above that are LOW-POWER (consistent but not constraining) and reported as such rather than counted as agreement. The analog-versus-stack variance recovery at matched budget is Figure 5.

Cell (SZA / nm)	Estimator/referee	Band/referee	Verdict
101° / 450	$1.06 \pm 0.04\times$	0.28	consistent
101° / 550	$0.87 \pm 0.05\times$	0.29	consistent
101° / 650	$0.92 \pm 0.05\times$	0.30	consistent
103° / 450	$0.94 \pm 0.05\times$	0.33	consistent
103° / 550	$1.13 \pm 0.05\times$	0.38	consistent
103° / 650	$1.09 \pm 0.05\times$	0.40	consistent

variance is not a sufficient convergence diagnostic in this regime; an external referee, or at minimum a cross-estimator gate, is required. Second, population control and directional guiding are not substitutes: weight windows alone (measured in an intermediate ledger) left the coefficient of variation unchanged, because splitting cannot multiply paths that the direction sampler never generates; the directional component alone leaves the weight spread untreated. The measured collapse in the coefficient of variation required both components.

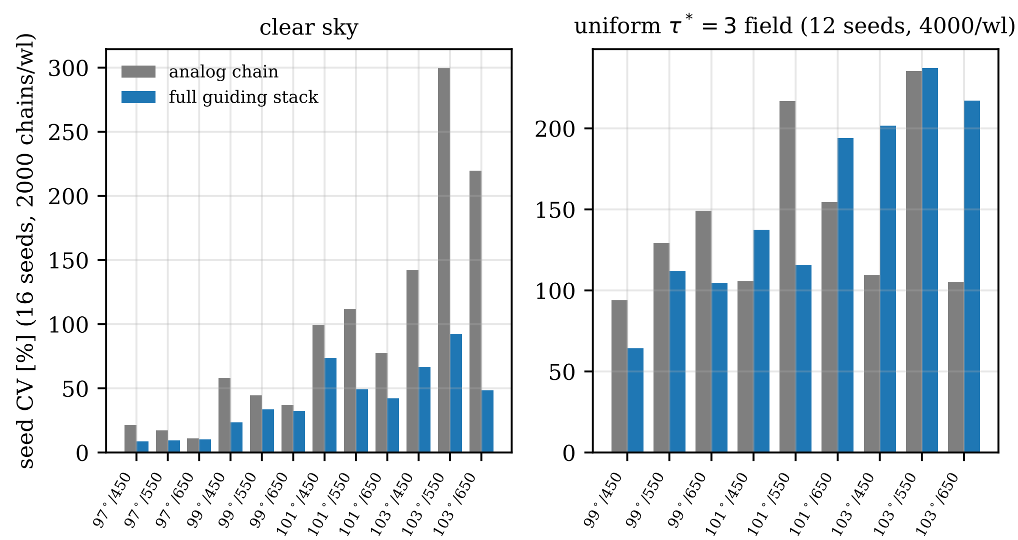

Figure 5: Measured effect of the guiding stack on the seed coefficient of variation. The left panel shows clear sky at the dawn-detection depressions (16 seeds of 2 000 chains per wavelength; labels give SZA/nm); the right panel shows the uniform $\tau^* = 3$ cloud field (12 seeds of 4 000 chains per wavelength). Gray bars show the analog polarized chain before this work; blue bars show the full stack (forced null-collision flights, VSPG, Dwivedi MIS, and weight windows).

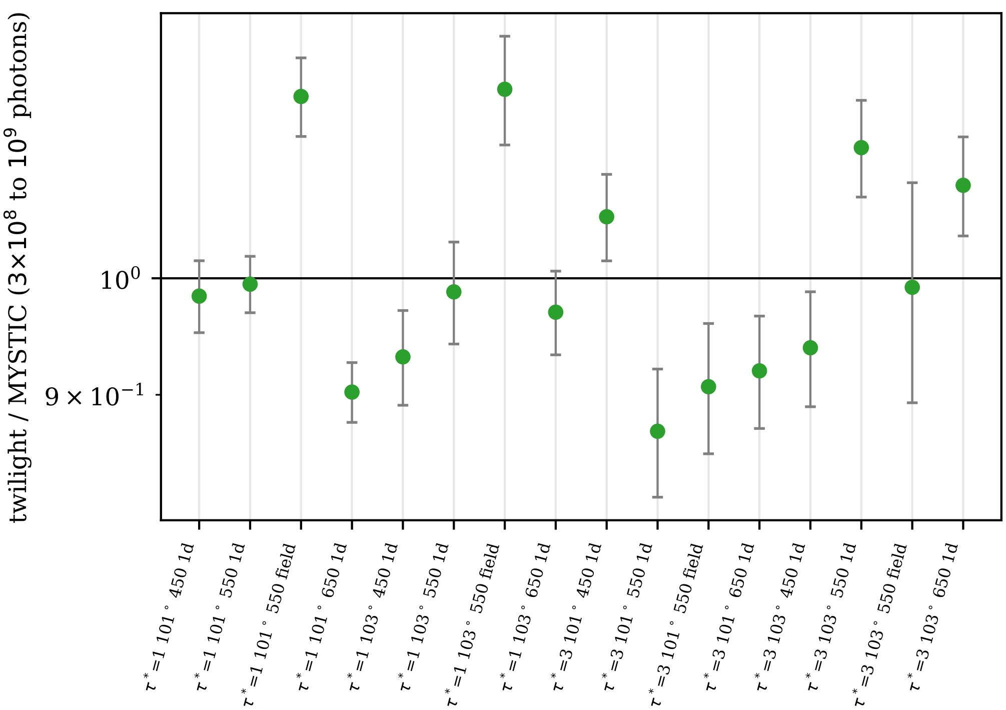

Figure 6: Every cell of the deep-regime referee table, shown as the model-to-MYSTIC ratio with seed standard errors; referee budgets are 3×10^8 to 10^9 photons. All sixteen cells, the one-dimensional and the three-dimensional field paths alike, are gated and statistically consistent (green; ratios 0.87 to 1.19). The field cells referee the voxel path on the equivalent horizontally uniform field of the same deck and share the one-dimensional rows' reference values; the two at SZA 103°, low-power at earlier seed budgets, close at 1024 and 512 seeds with the field estimator unchanged.

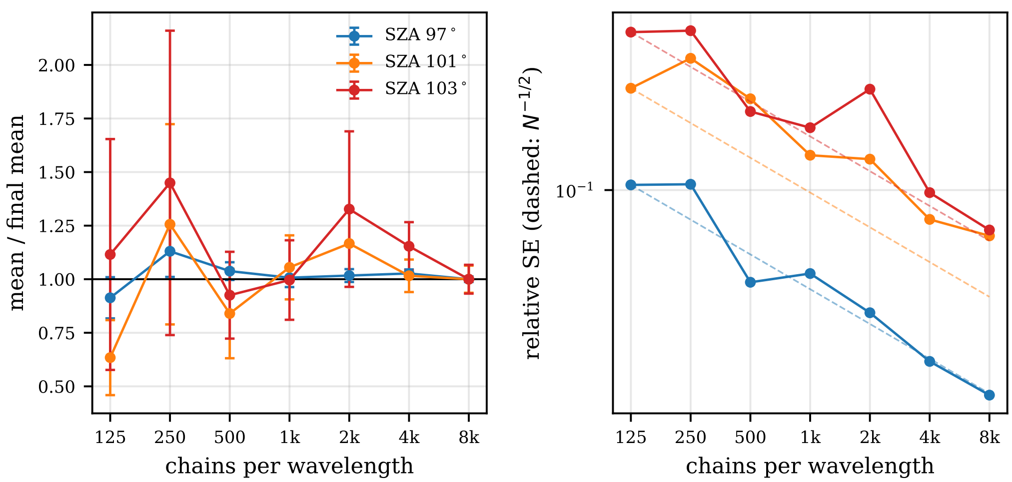

Figure 7: Convergence of the production polarized estimator with chain budget (clear sky, zenith view, 550 nm, SZA 97°, 101°, and 103°, 12 seeds per point). The left panel shows the running mean normalized to the largest budget, with seed standard errors; the right panel shows the relative standard error against the ideal $N^{-1/2}$ slope (dashed).

4. Independent implementations and parity methodology

The estimator is implemented four times: a scalar CPU chain, the production polarized CPU chain, a Metal shading language kernel, and a WGSL kernel that runs on any Vulkan, DirectX 12, or Metal adapter through the wgpu abstraction (this is the deployment path for headless Linux servers). The GPU kernels implement the full stack including weight-window splitting (a capped thread-local stack; the measured deep-cell effect is given in Section 7); buffer layouts are shared byte-for-byte between the two shader languages and pinned to the host by parse tests that read the shader source and compare every offset constant against the packing code.

Parity between implementations is enforced statistically. For each gate configuration (clear sky, uniform decks, checkerboard fields, a measured field) the CPU reference and the GPU kernel are run over K independent seeds; the acceptance band on the ratio of means is derived from the measured seed coefficients of variation before the comparison is made, and the derivation is quoted in the gate's documentation. Bands that cannot fail are treated as defects: an adversarial audit of the gate suite found two such vacuous bands (one absolute band wider than the measured means, one two-sided band wide enough to pass a 20% bias) and both were rewritten as ratio gates with bands that constrain. On current hardware the Metal and wgpu backends agree with the CPU to ratios of 0.94 to 1.04 on cloud fields and 0.995 to 1.010 on clear sky, with the GPU deep-field seed coefficient of variation now below the CPU's own (the GPU kernels carry the weight-window splitting and guiding stack; the forward-informed importance map remains CPU-only, so the CPU stays the formal variance reference). The bidirectional connection estimator of Section 3.5 likewise runs in the two CPU chains only; the GPU kernels retain the pre-connection estimator, whose expectation is identical, so the cross-implementation comparisons at deep solar zenith angles remain statistical same-mean parities rather than draw-for-draw identities, and the CPU chains carry the deep referee tier.

The portable WGSL path, which is the deployment route for non-Apple servers, has been exercised on discrete NVIDIA silicon through Vulkan rather than inferred from the Metal backend alone. On an NVIDIA A10G (driver reported through the Vulkan adapter) the full kernel suite passes (125 of 125 gates) and clear-sky parity holds to ratios of 0.9993, 0.9963, and 0.9957 at zenith angles of 95°, 97°, and 100°, matching the Apple figures to within their seed error. The adversarial fine-checkerboard field gate evaluates the

CPU and GPU means at full dispatch size and gives a ratio of 1.35 at SZA 97°; its acceptance band is wide (both estimators carry seed coefficients of variation near 0.26 in this deep broken-field regime), so the band is weak by construction, but it is derived from the measured seed dispersion rather than inflated to pass, and the ratio sits at the upper documented divergence of the GPU deep-field case, where the CPU-only importance map is what tightens the CPU reference (Section 7); the clear-sky and realistic-field paths, which are what a computed prayer day actually traverses, are the tight ones.

4.1. Computational cost

A full computed prayer day (the complete depression scan with the seven-patch detection fan, adaptive crossing refinement, and K-seed repeats) costs 237 s on an Apple M2 Pro GPU (Metal) and 1585 s on its eight performance cores (polarized CPU chains), with a scalar fast mode at 243 s; the numbers were measured under concurrent validation load and are upper bounds, and the wgpu backend runs the same WGSL on headless Vulkan hardware where dispatch shapes are not watchdog-limited: the full-size checkerboard field gate, which the display-driven Apple GPU can only bound from above before its watchdog intervenes, runs to completion in 544 s on the headless A10G. The referee side is far more expensive: the 10^9-photon MYSTIC cells of Section 5.2 cost hours each on the same machine, which is the practical argument for the estimator work of Section 3.

Beyond statistical parity, refactorings are pinned by bit-identity harnesses: a dump of radiance bit patterns across a matrix of configurations (clear sky, decks, fields, both chain families, multiple scattering modes, a range of SZA) is compared between trees, and every surface not intentionally changed must reproduce bit-for-bit. The harness has caught, among other things, a random-number-stream perturbation hidden behind an apparently neutral refactor. In the most recent campaign, 984 of 984 unchanged rows were bit-identical, while the one intentionally changed surface moved in all 82 of its rows.

5. Validation

The validation program is organized in three rings, from internal exactness through external reference codes to human ground truth. All referee caches (reference-code outputs, campaign run logs) are committed to the repository, so every number below can be re-derived without rebuilding the reference codes.

5.1. Ring 1: internal exactness

Analytic unit gates verify each estimator identity in isolation: the forced-flight first-arrival law against its closed form (per-bin agreement at five standard errors across a fine checkerboard, under tight and inflated majorants); majorant invariance (the estimate may not depend on the looseness of the bound); the exact equivalence of a uniform voxel field and the one-dimensional deck chains (0.04% at matched seeds, and ratio 0.999 to 1.000 in the dedicated deep-regime gate at SZA 101° and 103°); the telescoping normalization of the VSPG segment set ($p_{\mathrm{sum}} \equiv 1 - e^{-\tau_{\max}}$ to 10^{-12} on 200-traversal ducting paths); and the population-control identities. Physics-constraint gates check monotonicity (radiance decreasing along SZA ladders; the non-trivial ordering of clear versus thin-deck versus thick-deck red-channel zenith radiance, where thin decks brighten the red twilight zenith, a referee-corroborated finding) and smoothness (a χ^2 test across the forced-mode activation seam, which yields 11.6 on six degrees of freedom in the final configuration).

5.2. Ring 2: reference codes and measured skies

Table 2 summarizes the reference-code ring; Figures 8 and 9 show the clear-sky ratio bank and the cloud-slab scatter point by point, Figure 10 the true-3D cube comparison against SHDOM, and Figure 11 the absolute comparison against the measured Paranal twilight decay in both photometric bands.

Three points deserve comment. First, the absolute-scale DISORT comparison (no normalization of any kind) is the anchor of the radiometric chain: ratios 0.975 to 0.993 across SZA 60° to 85° bound any calibration error of the solar irradiance, cross sections, and quadrature at the percent level. Second, the cloud-slab bank (144 points across three optical depths, both of the code's independent Monte Carlo estimators, against both DISORT and MYSTIC on the identical delta-scaled Henyey-Greenstein problem) validates the cloud channel where the references are strong, so that the twilight extension rests only on geometry already validated in clear sky. Third, the ultra-deep rows extend external refereeing to SZA 103°, which to our knowledge is deeper than any published cloud-deck comparison (the complete cell-by-cell table is given in Appendix Appendix A). The two three-dimensional-field cells at

Table 2: Transport validation against reference codes and measured skies. SE denotes standard error. The deep-regime rows are refereed by dedicated MYSTIC runs of 3×10^8 to 10^9 photons with standard errors of 5.9 to 10.7%; the verdict taxonomy of Section 3.6 reports any non-constraining cell as LOW-POWER rather than as agreement, and no cell of the final campaign carries that label.

Reference	Regime	Agreement
DISORT, pseudospherical, 16 streams [31]	clear sky, SZA 60 to 90°	2.6 to 3.5% median (spectral shape)
DISORT, absolute scale	clear sky, SZA 60 to 85°	ratio 0.975 to 0.993
MYSTIC, backward spherical MC [23, 9]	clear sky, SZA 95 to 100°, 10^8 photons	1 to 14%
DISORT + MYSTIC, cloud slab	$\tau^* \in \{1, 3, 10\}$, SZA 30 to 60°	144/144 points within 3% + 2 SE, both estimators
SHDOM, true-3D cubes [10]	checkerboard and broken deck	64/64 and 48/48 points in band
MYSTIC, ultra-deep decks	τ^* 1 and 3, SZA 101 to 103°, up to 10^9 photons	all 16 cells, 1d and 3D-field, gated and consistent (ratios 0.87 to 1.19); the field cells closed by seed statistics at 1024 and 512 seeds
Measured zenith twilight decay [27, 19]	civil through astronomical twilight	1.2 to 5.5% per magnitude of decay

SZA 103° were for some time the statistically weak rows of this tier; larger seed campaigns (1024 seeds at $\tau^* = 1$, 512 at $\tau^* = 3$, of 16 000 photons each) closed them with the field estimator unchanged, confirming that the earlier low-power verdicts measured the seed budget rather than the estimator.

A note on reading ratios in the heavy-tailed regime is required, because the deep rows can mislead in both directions. When the seed distribution is dominated by rare bright chains, the sample mean at affordable budgets sits below the true mean with high probability (the typical seed misses the tail), so a ratio of, say, 0.6 with a seed coefficient of variation near 100% is what an unbiased estimator is expected to show; it is not evidence of bias. Conversely, a tight seed spread proves nothing if no seed contains the compensating

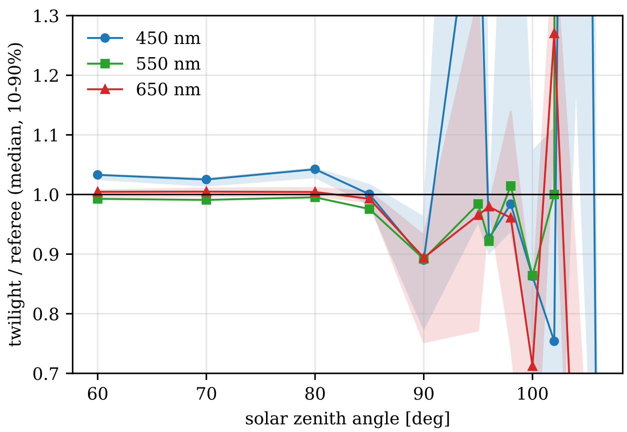

Figure 8: The clear-sky model-to-referee radiance ratio versus SZA at three wavelengths: the median and the 10 to 90% spread over all view geometries of the comparison bank (14 486 points) are shown. The referee is DISORT (pseudospherical) through SZA 90° and MYSTIC beyond.

chains, as the analog before bars of Figure 5 demonstrate. Our acceptance bands therefore combine the referee's standard error with the model-side seed spread, and rows whose bands cannot constrain a meaningful bias are labeled LOW-POWER rather than counted as agreement; the verdict taxonomy is applied mechanically, before the numbers are seen.

A pass count alone would overstate what the deep tier settles, so we report which side of the comparison limits it. Decomposing the two terms of the acceptance band across all sixteen cells, the model's relative seed standard error runs from 2.6 to 5.9 percent (one three-dimensional field cell reaches 9.9), while the referee's runs from 5.9 to 10.7 percent. *The referee is the larger term in fifteen of the sixteen cells.* The bands are consequently 25 to 46 percent wide, and driving the model-side error to zero everywhere would narrow them only to 23 to 37 percent. Additional model seeds therefore cannot sharpen this comparison materially; a sharper reference would.

Two consequences follow, and we state both rather than the pass count alone. First, the deep tier establishes agreement at the tens-of-percent level, a materially weaker claim than the percent-level DISORT anchor of Section 5.1, and the weakness is a property of the available references at these solar zenith angles rather than of the model. Second, the reported referee uncertainties are themselves worth testing. Pooling the eight fresh-seed replicas (two per

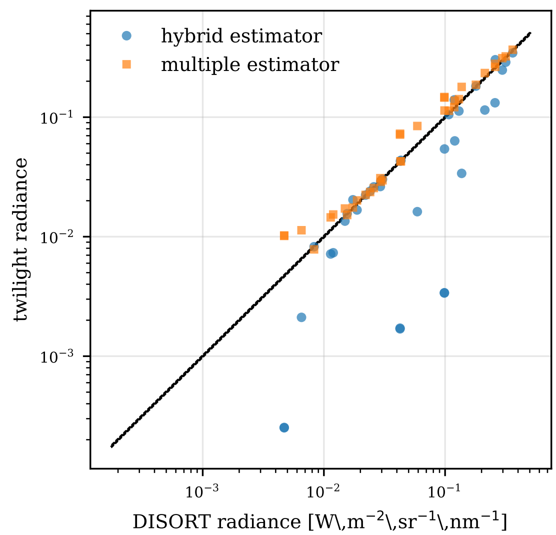

Figure 9: The cloud-slab bank compares both of the code's independent Monte Carlo estimators with DISORT on the identical delta-scaled Henyey-Greenstein problem (144 points across $\tau^* \in \{1, 3, 10\}$ and SZA 30 to 60°, spanning three decades of radiance). Dashed lines mark ±3%.

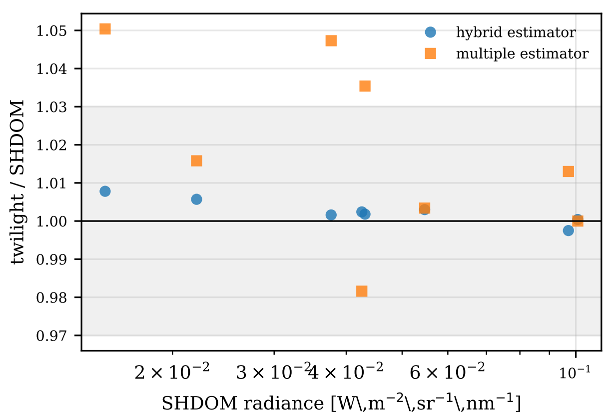

Figure 10: The true-3D cloud cube comparison against SHDOM is shown (checkerboard and broken-deck fields, both Monte Carlo estimators). The shaded band marks ±3%.

reference, run at each reference's own budget: 3×10^8 photons for the two SZA 101° references, 10^9 for the two at SZA 103°) by inverse variance tightens the four replicated $\tau^* = 3$ references (450 and 550 nm at SZA 101°; 550 and

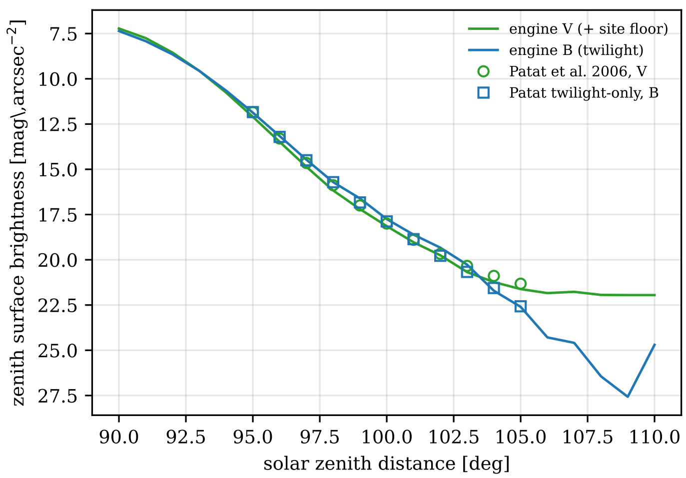

Figure 11: The absolute zenith twilight decay is compared with the measured Paranal data of Patat et al. [27]: the engine V band (with the site's measured night-sky floor added) and the twilight-only B band, in magnitudes per square arcsecond, against the published quadratic fits evaluated at integer zenith distances. Over ten magnitudes of decay the engine tracks the measurement to 0.1 to 0.4 mag.

650 nm at SZA 103°) from 6.1–10.6 to 3.6–5.7 percent relative standard error, narrowing the six affected tier rows (the 550 nm references are shared between the one-dimensional and field rows) by 24 percent in band, with every verdict unchanged and four of the six ratios moving toward unity. The same pooling permits a consistency test of the quoted sigmas: three of the four cells give χ^2 per degree of freedom between 0.29 and 0.63, but the deepest and reddest cell ($\tau^* = 3$, SZA 103°, 650 nm) gives 4.56, meaning three independent estimates of one quantity disagree about 2.1 times more than their own error bars allow. That is the cell where the heavy tail is expected to defeat the central-limit assumption behind a Monte Carlo standard error, and it is a caution against reading the 3σ Gaussian bands as if the underlying distribution were established as normal.

5.3. Ring 3: human observation campaigns

The application layer (described fully in a companion paper) converts simulated radiance into detection times through a psychophysical criterion. In brief: for each scanned depression the model simulates five band patches across the dawn arch and two dark reference patches; patch i is distinct when its growth above its own deep-night baseline exceeds the threshold set by the adaptation luminance,

$$L_i(\theta) - B_i \geq f\, C_{\mathrm{thr}}(L_{\mathrm{ref}})\, L_{\mathrm{ref}}, \tag{1}$$

with C_{thr} the laboratory threshold-versus-intensity function and f a single calibrated field factor; dawn is declared when the test passes across the band's lateral extent. In detail: seven simulated sky patches (five across the dawn band, two dark references), Weber contrast of each band patch's growth above its own deep-night baseline against threshold-versus-intensity data [4, 7], with a single calibrated field factor bridging laboratory detection of sharp-edged stimuli to field detection of the borderless dawn gradient. For the purposes of the present paper, the campaigns function as an end-to-end validation of the radiance model: if the simulated absolute radiance, its spectral shape, and its spatial gradient across the dawn band were wrong, no single constant could reproduce detection times across sites, seasons, and latitudes.

Calibrated on the desert campaigns that report a documented central-mean dawn, the model reproduces sixteen published visual and instrumental campaigns, from a 1909 photographic campaign at Aswan [24] through the modern KACST desert program [1], the Hail campaign [18], and a calibrated-camera study [22], with a median absolute residual of 0.26° of solar depression over the fourteen scored rows. The decisive out-of-sample test is the OpenFajr program in Birmingham, UK [25]: 42 dated all-sky-camera series judged by a 19-member observer panel across a full year at 52.4°N. The model, with zero retuning, reproduces the panel's summer plateau (about 12.6°, deepening to 11.9° at the solstice, against 12.3 to 12.7° observed) and the double-peaked seasonal curve, including the summer relaxation that UK observers apply as an empirical rule; in the model it emerges from persistent-twilight geometry, some twenty degrees of latitude from the desert calibration campaigns.

The calibrated constant is fixed, and stress-tested, by inversion. The published campaigns do not report a uniform statistic: several headline depressions are documented upper bounds (mean or median plus one to two standard deviations, the observing groups' stated "highest value of confidence"), not central means [15, 16]. Restricted to the campaigns that report a genuine central mean – Riyadh (14.58°), Aswan (14.90°), and Kottamia (14.665°) – the implied factor is 53, 51, and 57 (agreeing to within four units of f), an optimum of $f = 53$ with a plateau to 60; the production constant is set to 56, within that band. Hail's central mean (14.01°) is a half-degree shallower and implies $f \approx 81$; an earlier three-site cluster gave that lone anomaly a third of its weight and was pulled to $f = 70$. The remaining campaigns' headline values are upper bounds whose raw central means (Tubruq 13.14°, Assiut 11.25°) are far shallower and reflect all-nights or light-polluted averages the clear-sky model does not represent; that the model at $f = 56$ reproduces the upper bounds is expected, since a site's mean + 2SD dawn is its deepest, clear-sky detection. The inversion is latitude-independent.

6. Input uncertainty propagation

Because the intended output is a time, the model propagates its input uncertainties to a per-event standard deviation, reported alongside every computed time. Three background terms are propagated through the measured sensitivity of the detection margin at the crossing: the skyglow atlas calibration (0.30 in log radiance), the street-light duty cycle (a bimodal veil where part-night switching dominates the reference sky), and the aerosol input (either a climatological bracket or the per-value envelope of a measured product; the model ingests the Copernicus Atmosphere Monitoring Service (CAMS) aerosol optical depth as an excess over a baseline measured at the calibration sites). The realized sensitivities validate against dedicated bracket campaigns: 13 to 17 degrees of depression per unit excess aerosol optical depth in the linear regime, with the measured saturation behavior where the crossing slope steepens. Representative outputs: a desert site reports ±3.0 min; a mid-latitude winter city under sodium lighting reports ±7.2 min, dominated by the street-light duty-cycle term, which no public data feed currently supplies. We consider the width of these intervals informative rather than a defect: a deterministic minute would misrepresent what the inputs can support.

7. Accuracy boundaries and open extensions

Every boundary below is quantified, externally refereed where a referee exists, and maintained as a living section of the repository documentation; none affects the validated operating envelope of the preceding sections. (i) The three-dimensional field path retains the analog polarized chain: the Sun-side bidirectional connection estimator of Section 3.5, which removes the dependence of the deep signal on rare long lateral random walks, requires the one-dimensional combined channel and does not yet extend to heterogeneous voxel fields. The consequence is a convergence cost, not a bias. The history of the deep tier is instructive here: an eight-fold seed increase (to 1024 seeds) resolved every one-dimensional SZA 101° cell to referee consistency, including the 550 nm cell that at 128 seeds had read about forty percent low (under-sampling of the heavy tail, not a bias); the one-dimensional SZA 103° cells were then closed by the connection estimator; and the two three-dimensional field cells at SZA 103° ($\tau^* = 1$ and 3, 550 nm), low-power at earlier seed budgets, pass the 10^9-photon referee at 1024 and 512 seeds respectively with the field estimator unchanged. These field cells referee the voxel path on the equivalent horizontally uniform field of the same deck, against the same reference values as the corresponding one-dimensional rows; the SHDOM cube comparisons referee genuinely heterogeneous fields, but at moderate solar zenith angle, so an external deep-twilight referee on a heterogeneous scene remains open. The field path therefore converges at practical budgets only through large seed campaigns; extending the registry connections to voxel fields, which would restore the one-dimensional convergence rate, is the identified future work. Full bidirectional estimation had earlier been deferred because multiple importance sampling weights in a refracting spherical atmosphere carry a subtle-bias risk in precisely the regime hardest to referee; the per-path shared-weight construction, whose two weights sum to unity by evaluation of a single function, is what made the combination safe to deploy on the one-dimensional channel. (ii) Zenith views over broken cloud decks previously starved the importance sampler; a lateral-escape seed lobe (an azimuthally isotropic Dwivedi component folded into the seed mixture's balance-heuristic MIS weight, active only for cloud-coupled seeds under heterogeneous fields in the deep regime) closes the residual: a gate-measured +14% mean recovery with the seed coefficient of variation halved at SZA 100°, the restored zenith gate row passing two-sided, and bit-identity preserved on every other surface (these are internal gate diagnostics, since the analog reference here is noise-limited). The analog reference above SZA 100° over broken decks remains noise-limited at affordable budgets, so the configuration is refereed by the gate row rather than by analog parity. (iii) GPU kernels now carry the complete estimator stack including weight-window splitting (a four-slot thread-local stack with the CPU's roulette and split identities; measured on the uniform deck gate, the deep-cell GPU seed coefficient of variation falls from 0.175 to 0.136, below the CPU's 0.269, with a 10 to 13 percent wall-clock improvement as roulette kills outweigh split divergence); the forward-informed importance map and the deeper CPU stack remain CPU-only, so the CPU path stays the formal variance reference. (iv) Sea-horizon sites carry a documented open offset: a one-dimensional aerosol column cannot represent the directional marine boundary layer along the dawn path. The voxel framework already accepts three-dimensional aerosol through its gray extinction channel (a coastline-masked marine boundary-layer field built this way is under evaluation), and a spectral secondary aerosol grid would extend the per-shell majorants additively, leaving the null-collision classification unchanged. (v) The evening red-twilight (shafaq) calibration rests on instrumental rather than panel data, as no color-resolved modern campaign exists. (vi) The model's own field-instrument calibration campaign (a sky-quality-meter protocol included with the code) has not yet begun; all human-observation validation is against published campaigns.

8. Conclusions

We have described and validated an open-source backward Monte Carlo model of absolute spectral sky radiance through deep twilight, with spherical refracting geometry, full polarization, measured three-dimensional cloud fields, and a variance-reduction stack engineered against the heavy-tailed regime beyond SZA 100°. External refereeing now reaches SZA 103° on cloud decks at 10^9 reference photons, with every cell of the deep tier, one-dimensional and three-dimensional field paths alike, gated and consistent (ratios 0.87 to 1.19); absolute clear-sky radiance is anchored to DISORT at the percent level; and the end-to-end chain, radiance through psychophysics, reproduces sixteen human dawn-observation campaigns with a median absolute residual of 0.26° of depression over the fourteen scored rows, including a year-long out-of-sample panel study some twenty degrees of latitude from the desert calibration campaigns.

Beyond the specific results, we advocate two methodological practices in radiative transfer code development (the yield of the most recent review round is summarized in Table 3). First, pre-registered statistical acceptance bands, derived from measured variance before the comparison and audited for vacuousness, in place of informal agreement plots. Second, adversarial review of estimator mathematics as a routine phase: the compensating-error pairs that such review has twice found in this code (a sub-surface ground-bounce misclassification masked by near-cancellation; a silent importance-buffer truncation preserving total probability while redistributing it) are

Table 3: The sixteen findings of the most recent adversarial review round, by class, all closed before release. The compensating-error findings are the class that neither unit tests nor aggregate validation reliably catch.

Class	Count	Representative finding
Estimator (compensating-error)	3	importance-buffer overflow redistributing collision probability
Estimator (geometry/state)	3	detection fan frozen at the evening solar azimuth under three-dimensional fields
Photometry and inputs	5	photopic atlas value applied as a mesopic veil
Vacuous or mislabeled gates	3	absolute band wider than the measured means
Infrastructure	2	GPU seed salt discarded, collapsing repeat independence

precisely the class of defect that neither unit tests nor aggregate validation reliably catch.

The simulator, the application layer, the validation caches, and the scripts that regenerate every table in this paper are available in the public repository; a versioned release accompanies this paper.

Appendix A. The complete deep-regime referee table

Table A.4 lists every cell of the ultra-deep referee comparison of Section 5.2 with its budgets and verdicts, as archived with the code release.

This table was re-measured on the final estimator: all sixteen rows pass the gate. The two SZA 103° field rows, low-power at the earlier 128-seed budget, pass at 1024 and 512 seeds with the field estimator unchanged, and no row carries the false-precision signature of the analog estimator (the pre-stack field estimator at $\tau^* = 3$, SZA 103° sat at 0.13× with a deceptively tight spread; the guided stack replaces it with an honest, wider band). The one-dimensional cells of Table 1 remain the tightest power statements.

Table A.4: The complete deep-regime referee table, generated from the archived comparison data: the 1d rows and the $\tau^* = 1$ field rows at 1024 seeds of 16 000 photons per wavelength, the $\tau^* = 3$ field rows at 512 seeds. tw is the model mean, CV its per-seed coefficient of variation, and seeds the seed count; the MYSTIC column gives the referee estimate and its photon budget. Verdicts follow the taxonomy of Section 5.2. Radiances are in $\mathrm{W\,m^{-2}\,sr^{-1}\,nm^{-1}}$.

τ^*	SZA	nm	path	tw	CV [%]	seeds	MYSTIC	photons	ratio	verdict
1	101	450	1d	1.51×10^{-7}	104	1024	1.53×10^{-7}	3×10^8	0.98	PASS
1	101	550	1d	1.16×10^{-7}	82	1024	1.16×10^{-7}	3×10^8	0.99	PASS
1	101	550	field	1.37×10^{-7}	114	1024	1.16×10^{-7}	3×10^8	1.18	PASS
1	101	650	1d	7.92×10^{-8}	86	1024	8.78×10^{-8}	3×10^8	0.90	PASS
1	103	450	1d	2.55×10^{-8}	137	1024	2.73×10^{-8}	10^9	0.93	PASS
1	103	550	1d	1.93×10^{-8}	147	1024	1.96×10^{-8}	10^9	0.99	PASS
1	103	550	field	2.32×10^{-8}	156	1024	1.96×10^{-8}	10^9	1.19	PASS
1	103	650	1d	1.26×10^{-8}	121	1024	1.30×10^{-8}	10^9	0.97	PASS
3	101	450	1d	1.33×10^{-7}	125	1024	1.26×10^{-7}	3×10^8	1.06	PASS
3	101	550	1d	1.07×10^{-7}	185	1024	1.22×10^{-7}	3×10^8	0.87	PASS
3	101	550	field	1.11×10^{-7}	133	512	1.22×10^{-7}	3×10^8	0.91	PASS
3	101	650	1d	7.17×10^{-8}	162	1024	7.79×10^{-8}	3×10^8	0.92	PASS
3	103	450	1d	2.25×10^{-8}	166	1024	2.40×10^{-8}	10^9	0.94	PASS
3	103	550	1d	1.70×10^{-8}	140	1024	1.51×10^{-8}	10^9	1.13	PASS
3	103	550	field	1.50×10^{-8}	224	512	1.51×10^{-8}	10^9	0.99	PASS
3	103	650	1d	1.09×10^{-8}	143	1024	1.00×10^{-8}	10^9	1.09	PASS

Code and data availability

The source code is available at https://github.com/Muno459/twilight (dual-licensed MIT/Apache-2.0); release v1.0.0 corresponds to this paper. The release includes a validation archive containing the referee outputs, campaign run records, and the exact data behind every figure and table of this paper, together with the scripts that regenerate them. Reference-code comparisons used libRadtran 2.0.6 (DISORT, MYSTIC) and SHDOM. The three-dimensional cloud reconstruction uses the public cloud3d network [3].

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This research received no external funding. The author declares no competing interests.

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